The Artificial Neuron KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

Word bank: adds compares multiplies

1. Should I go? Answer 1 (yes) or 0 (no). Then choose how much each one matters to you (0 to 3).

	Answer (1 or 0)	**How much it matters**	**Answer × how much**
Is it sunny?	Answers vary.		
Are my friends going?			
Is my homework done?			
		Total	

Is the total 4 or more? _____ Do I go? _____ Did my partner decide the same? _____

2. Multiply, add, compare.

An artificial neuron multiplies each input by its weight, adds them up, and compares the total with the threshold.

Inputs: 1, 0, 1. Weights: 3, 2, −1. Threshold: 2.

Total = $(1 \times 3) + (0 \times 2) + (1 \times -1) =$ 2 Does it fire (2 or more)? yes

Inputs: 1, 1, 1. Weights: 1, −2, −1. Threshold: 2.

Total = $1 - 2 - 1 = -2$ Does it fire? no

3. Be a neuron. Weights: **A:** +2, **B:** +2, **C:** −1. Threshold: **3 or more**.

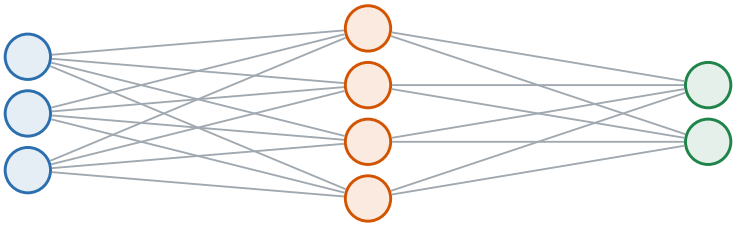
Round	**Inputs that are on**	**Total**	**Fire?**
1	A and B	4	yes
2	A and C	1	no
3	A, B, and C	3	yes
4	B and C	1	no
5	A only	2	no

Word bank: hidden input output threshold weight
You can use a word more than once.

4. Real neuron, artificial neuron. Match each part.

Real neuron	Artificial neuron
Dendrites take signals in	input
Strong or weak synapses	weight
Fires only if the total is big enough	threshold

5. From one neuron to a network. Name each layer.



input layer hidden layer output layer

Each line is a weight. How many lines? $3 \times 4 + 4 \times 2 =$ 20

Why is the middle layer called hidden? You don't see what those neurons decide.

6. Talk about it. ChatGPT has billions of weights. Nobody can choose them by hand.

I think an AI with billions of weights gets them by Answers vary. (Next lesson: it learns them.)

Go Further (Day 2). A neuron that sees. Write the weights that got the most test pictures right.

0	−1	**0**
−1	+3	−1
0	−1	**0**

Threshold: 0

Test pictures right: 6 out of 6

One good answer. Others work too.

Which squares got **positive** weights? Why? The center: it's black in an X, not in an O.

Which squares got **negative** weights? Why? The sides: they're black in an O, not in an X.